CORE PURE / PROOF BY INDUCTION

Proof by Induction - Series

Questions by topic 10 questions · 73 marks

Questions

Question 1	*6 marks*	2
Question 2	*6 marks*	3
Question 3	*6 marks*	4
Question 4	*6 marks*	5
Question 5	*12 marks*	6
Question 6	*8 marks*	7
Question 7	*6 marks*	8
Question 8	*6 marks*	9
Question 9	*11 marks*	10
Question 10	*6 marks*	11

1. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} r\,2^r = 2 + (n-1)2^{n+1}$$

[6]

2. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} (2^r + 2 \times 3^r) = 2^{n+1} + 3^{n+1} - 5$$

[6]

3. Prove by induction that for all positive integers n

$$\sum_{r=1}^{n} r(r+1)(r+2) = \frac{1}{4}n(n+1)(n+2)(n+3)$$

[6]

4. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^{n} r\, 3^{r-1} = \frac{1 + (2n-1)3^n}{4}$$ **[6]**

5. **(a)** Prove by induction that, for all integers $n \geq 1$

$$\sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n+1)^2$$

[6]

(b) Hence, or otherwise, write down a factorised expression for the sum of the first $2n$ cubes

$$1^3 + 2^3 + 3^3 + \cdots + (2n)^3$$

[1]

(c) Use the formula in part (a) to write down a factorised expression for the sum of the first n even cubes

$$2^3 + 4^3 + 6^3 + \cdots + (2n)^3$$

[2]

(d) Hence, or otherwise, show that

$$1^3 + 3^3 + 5^3 + \cdots + (2n-1)^3 = an^2(bn^2 - 1)$$

where a and b are rational numbers to be determined. **[3]**

6. **(a)** Prove by induction that, for all positive integers n

$$\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{2n+1}$$ **[6]**

(b) Hence, show that, for all positive integers n

$$\sum_{r=n+1}^{2n} \frac{1}{(2r-1)(2r+1)} = \frac{n}{(an+b)(cn+d)}$$

where a, b, c and d are integers to be determined. **[2]**

7. Prove by induction that, for $n \geq 1$

$$\sum_{r=1}^{n} \frac{r}{2^r} = 2 - \frac{n+2}{2^n}$$ **[6]**

8. Prove by induction that for $n \in \mathbb{Z}^+$

$$\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$ **[6]**

9. **(a)** Prove by induction that for all positive integers n

$$\sum_{r=1}^{n}(2r-1)^3 = n^2(2n^2-1)$$

[6]

(b) Given that

$$\sum_{r=1}^{2n}(2r-1)^3 = k\sum_{r=1}^{n}(2r-1)^3$$

show that

$$n^2 = \frac{k-4}{2(k-16)}$$

[5]

10. Prove by induction that for all positive integers n

$$\sum_{r=1}^{n} \log\left(1 - \frac{1}{4r^2}\right) = \log\left(\frac{(2n)!(2n+1)!}{2^{4n}(n!)^4}\right) \quad \textbf{[6]}$$